

which follows from the fact that $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$, and $\Gamma(1-s) = -s\Gamma(-s)$ (see Theorem 1.4 and Lemma 1.2 in Chapter 6). The combination of (13) and (14), together with the fact that for large s , $(1 + O(1/|s|^{1/2}))^{-1} = 1 + O(1/|s|^{1/2})$, allows us then to extend (11) to the whole sector S_δ , thereby completing the proof of the theorem.

3 The Airy function

The Airy function appeared first in optics, and more precisely, in the analysis of the intensity of light near a caustic; it was an important early instance in the study of asymptotics of integrals, and it continues to arise in a number of other problems. The **Airy function** Ai is defined by

$$(15) \quad \text{Ai}(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(x^3/3+sx)} dx, \quad \text{with } s \in \mathbb{R}.$$

Let us first see that because of the rapid oscillations of the integrand as $|x| \rightarrow \infty$, the integral converges and represents a continuous function of s . In fact, note that

$$\frac{1}{i(x^2 + s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) = e^{i(x^3/3+sx)},$$

so if $a \geq 2|s|^{1/2}$, we can write the integral $\int_a^R e^{i(x^3/3+sx)} dx$ as

$$(16) \quad \int_a^R \frac{1}{i(x^2 + s)} \frac{d}{dx} \left(e^{i(x^3/3+sx)} \right) dx.$$

We may now integrate by parts and let $R \rightarrow \infty$, to see that the integral converges uniformly, and that as a result $\int_a^\infty e^{i(x^3/3+sx)} dx$ is also continuous for $|s| \leq a^2/4$. The same argument works for the integral from $-\infty$ to $-a$ and our assertion regarding $\text{Ai}(s)$ is established.

A better insight into $\text{Ai}(s)$ is given by deforming the contour of integration in (15). A choice of an optimal contour will appear below, but for now let us notice that as soon as we replace the x -axis of integration in (15) by the parallel line $L_\delta = \{x + i\delta, x \in \mathbb{R}\}$, $\delta > 0$, matters improve dramatically.

In fact, we may apply the Cauchy theorem to $f(z) = e^{i(z^3/3+sz)}$ over the rectangle γ_R shown in Figure 3.

One observes that $f(z) = O(e^{-\delta x^2})$ on L_δ , while $f(z) = O(e^{-yR^2})$ on the vertical sides of the rectangle. Thus since $\int_0^\delta e^{-yR^2} dy \rightarrow 0$ as



Figure 3. The line L_δ and the contour γ_R

$R \rightarrow \infty$, we see that

$$\text{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} e^{i(z^3/3 + sz)} dz.$$

Now the majorization $f(z) = O(e^{-\delta x^2})$ continues to hold for each complex s , and hence because of the (rapid) convergence of the integral, $\text{Ai}(s)$ extends to an entire function of s .

We note next that $\text{Ai}(s)$ satisfies the differential equation

$$(17) \quad \text{Ai}''(s) = s\text{Ai}(s).$$

This simple and natural equation helps to explain the ubiquity of the Airy function. To prove (17) observe that

$$\text{Ai}''(s) - s\text{Ai}(s) = \frac{1}{2\pi} \int_{L_\delta} (-z^2 - s)e^{i(z^3/3 + sz)} dz.$$

But $-(z^2 + s)e^{i(z^3/3 + sz)} = i \frac{d}{dz}(e^{i(z^3/3 + sz)})$, so

$$\text{Ai}''(s) - s\text{Ai}(s) = \frac{i}{2\pi} \int_{L_\delta} \frac{d}{dz}(f(z)) dz = 0,$$

since $f(z) = e^{i(z^3/3 + sz)}$ vanishes as $|z| \rightarrow \infty$ along L_δ .

We now turn to our main problem, the asymptotics of $\text{Ai}(s)$ for large (real) values of s . The differential equation (17) shows us that we may expect different behaviors of the Airy function when $|s|$ is large, depending on whether s is positive or negative. To see this, we compare the equation with a simple analogue

$$(18) \quad y''(s) = Ay(s),$$

where A is a large constant, with A positive when considering s positive and A negative in the other case. The solutions of (18) are of course $e^{\sqrt{A}s}$ and $e^{-\sqrt{A}s}$, the first growing rapidly, and the second decreasing rapidly as $s \rightarrow \infty$, if $A > 0$. A glance at the integration by parts following (16) shows that $\text{Ai}(s)$ remains bounded when $s \rightarrow \infty$. So the comparison with $e^{\sqrt{A}s}$ must be dismissed, and we might reasonably guess that $\text{Ai}(s)$ is rapidly decreasing in this case. When $s < 0$ we take $A < 0$ in (18). The exponentials $e^{\sqrt{A}s}$ and $e^{-\sqrt{A}s}$ are now oscillating, and we can therefore presume that $\text{Ai}(s)$ should have an oscillatory character as $s \rightarrow -\infty$.

Theorem 3.1 *Suppose $u > 0$. Then as $u \rightarrow \infty$,*

$$(i) \quad \text{Ai}(-u) = \pi^{-1/2} u^{-1/4} \cos\left(\frac{2}{3}u^{3/2} - \frac{\pi}{4}\right)(1 + O(1/u^{3/4})).$$

$$(ii) \quad \text{Ai}(u) = \frac{1}{2\pi^{1/2}} u^{-1/4} e^{-\frac{2}{3}u^{3/2}} (1 + O(1/u^{3/4})).$$

To consider the first case, we make the change of variables $x \mapsto u^{1/2}x$ in the defining integral with $s = -u$. This gives

$$\text{Ai}(-u) = u^{1/2} I_-(u^{3/2}),$$

where

$$(19) \quad I_-(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{it(x^3/3-x)} dx.$$

Now write

$$I_-(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-s\Phi(x)} dx,$$

where $\Phi(x) = \Phi_-(x) = x^3/3 - x$, and we shall apply Proposition 2.2, which in this case, since s is purely imaginary, is the method of stationary phase. Note that $\Phi'(x) = x^2 - 1$, so there are two critical points, $x_0 = \pm 1$; observe that $\Phi''(x) = 2x$; also $\Phi(\pm 1) = \mp 2/3$.

We break up the range of integration in (19) into two intervals $[-2, 0]$ and $[0, 2]$ each containing one critical point, and two complementary integrals, $(-\infty, -2]$ and $[2, \infty)$.

Now we apply Proposition 2.2 to the interval $[0, 2]$ with $s = -it$, $x_0 = 1$, $\psi = 1/2\pi$, $\Phi(1) = -2/3$, $\Phi''(1) = 2$, and get a contribution of

$$\frac{1}{2\sqrt{\pi}} e^{-i\frac{2}{3}t} \left(\frac{1}{(-it)^{1/2}} + O\left(\frac{1}{|t|}\right) \right),$$

in view of (8). Similarly the integral over $[-2, 0]$ contributes

$$\frac{1}{2\sqrt{\pi}} e^{i\frac{2}{3}t} \left(\frac{1}{(it)^{1/2}} + O\left(\frac{1}{|t|}\right) \right).$$

Finally, consider the complementary integrals. The first is

$$\int_{-\infty}^{-2} e^{it\Phi(x)} dx = \lim_{N \rightarrow \infty} \int_{-N}^{-2} e^{it\Phi(x)} dx = \lim_{N \rightarrow \infty} \frac{1}{it} \int_{-N}^{-2} \frac{d}{dx} (e^{it\Phi(x)}) \frac{dx}{\Phi'(x)},$$

where $\Phi'(x) = x^2 - 1$. So an integration by parts shows that this is $O(1/|t|)$. The integral over $[2, \infty)$ is treated similarly. Adding these four contributions, and inserting them in the identity $\text{Ai}(-u) = u^{1/2} I_-(u^{3/2})$, proves conclusion (i) of the theorem.⁶

To deal with the conclusion (ii) of the theorem, we make the change of variables $x \mapsto u^{1/2}x$ in the integral (15), with $s = u$. This gives us, for $u > 0$,

$$\text{Ai}(u) = u^{1/2} I_+(u^{3/2}),$$

where

$$(20) \quad I_+(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-sF(x)} dx$$

and $F(x) = -i(x^3/3 + x)$. Now when $s \rightarrow \infty$, the integrand in (20) again oscillates rapidly, but here in distinction to the previous case, there is no critical point on the real axis, since the derivative of $x^3/3 + x$ does not vanish. A repeated integration by parts argument (such as we have used before) shows that actually the integral $I_+(s)$ has fast decay as $s \rightarrow \infty$. But what is the exact nature and order of this decrease? To answer this question, we would have to take into account the precise cancellations inherent in (20), and doing this by the above method does not seem feasible.

A better way is to follow the guiding principle used in the asymptotics of the Bessel function, and to deform the line of integration in (20) to a contour on which the imaginary part of $F(z)$ vanishes; having done this, one might then hope to apply Laplace's method, Proposition 2.1, to find the true asymptotic behavior of $I_+(s)$, as $s \rightarrow \infty$.

We describe the idea in the more general situation in which we assume only that $F(z)$ is holomorphic. To follow the approach suggested, we seek a contour Γ so that:

⁶An alternative derivation of this conclusion can be given as a consequence of the relation of the Airy function with the Bessel functions. See Problem 3 below.

- (a) $\text{Im}(F) = 0$ on Γ .
- (b) $\text{Re}(F)$ has a minimum on Γ at some point z_0 , and this function is non-degenerate in the sense that the second derivative of $\text{Re}(F)$ along Γ is strictly positive at z_0 .

Conditions (a) and (b) imply of course that $F'(z_0) = 0$. If as above, $F''(z_0) \neq 0$, then there are two curves Γ_1 and Γ_2 passing through z_0 which are orthogonal, so that $F|_{\Gamma_i}$ is real for $i = 1, 2$, with $\text{Re}(F)$ restricted to Γ_1 having a minimum at z_0 ; and $\text{Re}(F)$ restricted to Γ_2 having a maximum at z_0 (see Exercise 2 in Chapter 8). We therefore try to deform our original contour of integration to $\Gamma = \Gamma_1$. This approach is usually referred to as the method of **steepest descent**, because at z_0 the function $-\text{Re}(F(z))$ has a saddle point, and starting at this point and following the path of Γ_1 , one has the greatest decrease of this function.

Let us return to our special case, $F(z) = -i(z^3/3 + z)$. We note that

$$\begin{cases} \text{Re}(F) &= x^2y - y^3/3 + y, \\ \text{Im}(F) &= -x^3/3 + xy^2 - x. \end{cases}$$

We observe also that $F'(z) = -i(z^2 + 1)$, so we have two non-real critical points $z_0 = \pm i$ at which $F'(z_0) = 0$. If we choose $z_0 = i$, then the two curves passing through this point where $\text{Im}(F) = 0$ are

$$\Gamma_1 = \{(x, y) : y^2 = x^2/3 + 1\} \quad \text{and} \quad \Gamma_2 = \{(x, y) : x = 0\}.$$

On Γ_2 , the function $\text{Re}(F)$ clearly has a maximum at the point $z_0 = i$, and so we reject this curve. We choose $\Gamma = \Gamma_1$, which is a branch of a hyperbola, and which can be written as $y = (x^2/3 + 1)^{1/2}$; it is asymptotic to the rays $z = re^{i\pi/6}$, and $z = re^{i5\pi/6}$ at infinity. See Figure 4.

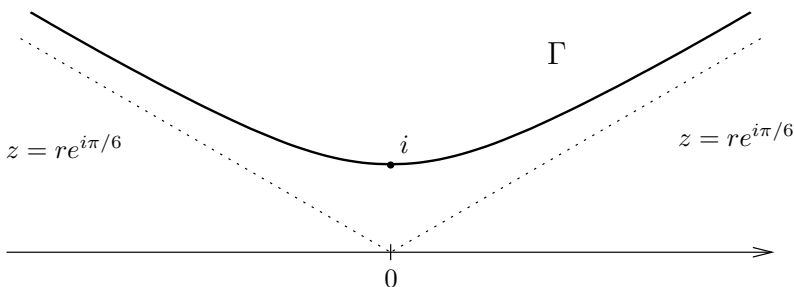


Figure 4. The curve of steepest descent

Next, we see that

$$(21) \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-sF(x)} dx = \frac{1}{2\pi} \int_{\Gamma} e^{-sF(z)} dz.$$

This identity is justified by applying the Cauchy theorem to $e^{-sF(z)}$ on the contour Γ_R that consists of four arc segments: the parts of the real axis and Γ that lie inside the circle of radius R , and the two arcs of this circle joining the axis with Γ . Since in this region $e^{-sF(z)} = O(e^{-cyx^2})$ as $x \rightarrow \pm\infty$, the contributions of the two arcs of the circle are $O(\int_0^\pi e^{-cR^2 \sin \theta} d\theta) = O(1/R)$, and letting $R \rightarrow \infty$ establishes (21).

We now observe that on Γ

$$\Phi(x) = \operatorname{Re}(F) = y(x^2 - y^2/3 + 1) = \left(\frac{8}{9}x^2 + \frac{2}{3}\right)(x^2/3 + 1)^{1/2},$$

since $y^2 = x^2/3 + 1$ there. Also, on Γ we have that $dz = dx + i dy = dx + i(x/3)(x^2/3 + 1)^{-1/2}dx$. Thus,

$$(22) \quad \frac{1}{2\pi} \int_{\Gamma} e^{-sF(z)} dz = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-s\Phi(x)} dx,$$

in view of the fact that $\Phi(x)$ is even, while $x(x^2/3 + 1)^{-1/2}$ is odd.

We note next that since $(1+u)^{1/2} = 1 + u/2 + O(u^2)$ as $u \rightarrow 0$,

$$\Phi(x) = \left(\frac{8}{9}x^2 + \frac{2}{3}\right) + \frac{2}{3} \frac{1}{2} \frac{x^2}{3} + O(x^4) = x^2 + \frac{2}{3} + O(x^4),$$

and so $\Phi''(0) = 2$. We now apply Proposition 2.1 to estimate the main part of the right-hand side of (22), by

$$\frac{1}{2\pi} \int_{-c}^c e^{-s\Phi(x)} dx,$$

where c is a small positive constant. Since $\Phi(0) = 2/3$, $\Phi''(0) = 2$, and $\psi(0) = 1/2\pi$, we obtain that this term contributes

$$e^{-\frac{2}{3}s} \left[\frac{1}{2\pi^{1/2}} \frac{1}{s^{1/2}} + O\left(\frac{1}{s}\right) \right].$$

The term $\int_c^\infty e^{-s\Phi(x)} dx$ is dominated by $e^{-2s/3} \int_c^\infty e^{-c_1 s x^2} dx$, which is $O(e^{-2s/3} e^{-\delta s})$ for some $\delta > 0$, as soon as $c > 0$. A similar estimate holds for $\int_{-\infty}^{-c} e^{-s\Phi(x)} dx$. Altogether, then,

$$I_+(s) = e^{-\frac{2}{3}s} \left[\frac{1}{2\pi^{1/2}} \frac{1}{s^{1/2}} + O\left(\frac{1}{s}\right) \right] \quad \text{as } s \rightarrow \infty,$$

and this gives the desired asymptotic (ii) for the Airy function.

4 The partition function

Our last illustration of the techniques developed in this appendix is in their application to the partition function $p(n)$, which was discussed in Chapter 10. We derive for it the main term of the remarkable asymptotic formula of Hardy-Ramanujan.

Theorem 4.1 *If p denotes the partition function, then*

$$(i) \quad p(n) \sim \frac{1}{4\sqrt{3}n} e^{Kn^{1/2}} \text{ as } n \rightarrow \infty, \text{ where } K = \pi\sqrt{\frac{2}{3}}.$$

(ii) *A much more precise assertion is that*

$$p(n) = \frac{1}{2\pi\sqrt{2}} \frac{d}{dn} \left(\frac{e^{K(n-\frac{1}{24})^{1/2}}}{(n-\frac{1}{24})^{1/2}} \right) + O(e^{\frac{K}{2}n^{1/2}}).$$

Note. Observe that $(n - \frac{1}{24})^{1/2} - n^{1/2} = O(n^{-1/2})$, by the mean-value theorem; hence $e^{K(n-\frac{1}{24})^{1/2}} = e^{Kn^{1/2}}(1 + O(n^{-1/2}))$, thus $e^{K(n-\frac{1}{24})^{1/2}} \sim e^{Kn^{1/2}}$, as $n \rightarrow \infty$. Of course, clearly $(n - \frac{1}{24})^{1/2} \sim n^{1/2}$, and in particular (ii) implies (i).

We shall discuss first, in a more general setting, how we might derive the asymptotic behavior of a sequence $\{F_n\}$ from the analytic properties of its generating function $F(w) = \sum_{n=0}^{\infty} F_n w^n$. Assuming for the sake of simplicity that $\sum F_n w^n$ has the unit disc as its disc of convergence, we can set forth the following heuristic principle: the asymptotic behavior of F_n is determined by the location and nature of the “singularities” of F on the unit circle, and the contribution to the asymptotic formula due to each singularity corresponds in magnitude to the “order” of that singularity.

A very simple example in which this principle is unambiguous and can be verified occurs when F is meromorphic in a larger disc, but has only one singularity on the circle, a pole of order r at the point $w = 1$. Then there is a polynomial P of degree $r - 1$ so that $F_n = P(n) + O(e^{-\epsilon n})$ as $n \rightarrow \infty$, for some $\epsilon > 0$. In fact, $\sum_{n=0}^{\infty} P(n)w^n$ is a good approximation to $F(w)$ near $w = 1$; it is the principal part of the pole of F . (See also Problem 4.)

For the partition function the analysis is not as simple as this example, but the principle stated above is still applicable when properly interpreted. To this task we now turn.

We recall the formula

$$\sum_{n=0}^{\infty} p(n) w^n = \prod_{n=1}^{\infty} \frac{1}{1 - w^n},$$